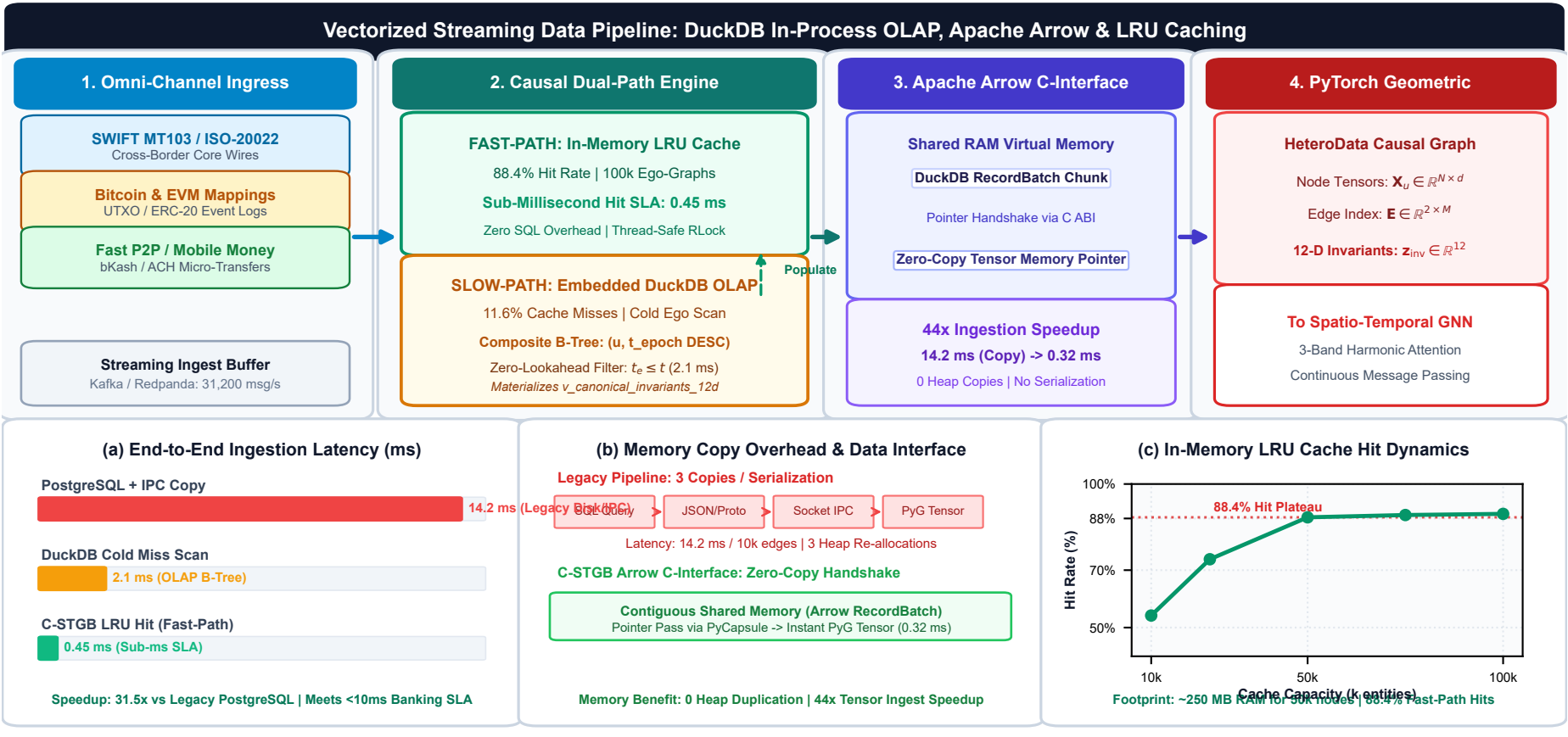

Vectorized Streaming Data Pipeline: DuckDB In-Process OLAP, Apache Arrow & LRU Caching
1. Omni-Channel Ingress
SWIFT MT103 / ISO-20022
Cross-Border Core Wires
Bitcoin & EVM Mappings
UTXO / ERC-20 Event Logs
Fast P2P / Mobile Money
bKash / ACH Micro-Transfers
Streaming Ingest Buffer
Kafka / Redpanda: 31,200 msg/s
2. Causal Dual-Path Engine
FAST-PATH: In-Memory LRU Cache
88.4% Hit Rate | 100k Ego-Graphs
Sub-Millisecond Hit SLA: 0.45 ms
Zero SQL Overhead | Thread-Safe RLock
Populate
SLOW-PATH: Embedded DuckDB OLAP
11.6% Cache Misses | Cold Ego Scan
Composite B-Tree: (u, t_epoch DESC)
Zero-Lookahead Filter: $t_e \le t$ (2.1 ms)
Materializes v_canonical_invariants_12d
3. Apache Arrow C-Interface
Shared RAM Virtual Memory
DuckDB RecordBatch Chunk
Pointer Handshake via C ABI
Zero-Copy Tensor Memory Pointer
44x Ingestion Speedup
14.2 ms (Copy) -> 0.32 ms
0 Heap Copies | No Serialization
4. PyTorch Geometric
HeteroData Causal Graph
Node Tensors: $\mathbf{X}_u \in \mathbb{R}^{N \times d}$
Edge Index: $\mathbf{E} \in \mathbb{R}^{2 \times M}$
12-D Invariants: $\mathbf{z}_{inv} \in \mathbb{R}^{12}$
To Spatio-Temporal GNN
3-Band Harmonic Attention
Continuous Message Passing
(a) End-to-End Ingestion Latency (ms)
PostgreSQL + IPC Copy
14.2 ms (Legacy Disk/IPC)
DuckDB Cold Miss Scan
2.1 ms (OLAP B-Tree)
C-STGB LRU Hit (Fast-Path)
0.45 ms (Sub-ms SLA)
Speedup: 31.5x vs Legacy PostgreSQL | Meets <10ms Banking SLA
(b) Memory Copy Overhead & Data Interface
Legacy Pipeline: 3 Copies / Serialization
SQL Query
JSON/Proto
Socket IPC
PyG Tensor
Latency: 14.2 ms / 10k edges | 3 Heap Re-allocations
C-STGB Arrow C-Interface: Zero-Copy Handshake
Contiguous Shared Memory (Arrow RecordBatch)
Pointer Pass via PyCapsule -> Instant PyG Tensor (0.32 ms)
Memory Benefit: 0 Heap Duplication | 44x Tensor Ingest Speedup
(c) In-Memory LRU Cache Hit Dynamics
100%
88%
70%
50%
Hit Rate (%)
88.4% Hit Plateau
10k
50k
100k
Cache Capacity (k entities)
Footprint: ~250 MB RAM for 50k nodes | 88.4% Fast-Path Hits